Parameters and Calibration: The Numeric Foundation

Purpose

This document is the single source of truth for every numeric parameter in the simulation. When other documents reference a number, this is where the canonical value lives. Agents do not need to memorize these -- the orchestrator uses them internally -- but they provide grounding for what "reasonable" looks like.

Demand System Parameters

Multinomial Logit Utility

```
U_i = a * quality_i - b * price_i + g * brand_i + xi_i + eps_i
U_0 = V_0(t) + macro_shock + eps_0     (outside option)
```

Parameter	Symbol	Value	Units	Notes
Price coefficient	b	0.000015	per dollar	Higher price -> lowe
Quality coefficient	a	1.0	per quality point	Normalized to 1.0
Brand coefficient	g	0.4	per brand point	Brand matters, less
Outside option base	V_0(0)	3.5	utility units	High = most people c
Outside option decay	lambda_V0	0.03/quarter	per quarter	As awareness grows,
Outside option floor	V_0_min	0.5	utility units	Some people will nev
Logit scale paramete	mu	1.0	--	Standard normalizati
Macro demand shock s	sigma_macro	0.08	--	~+-16% in extreme qu
Firm taste shock std	sigma_xi	0.05	--	~+-10% firm-specific

Potential Market Size

Parameter	Value	Units
Base potential patients (Q1 2031)	600,000,000	adults 50+ in high-income countrie
Initial awareness rate	0.15	fraction who know SRT exists
Awareness growth (natural)	0.02/quarter	per quarter, additive
Awareness growth (per $1M SGA indu	0.001	per quarter per $1M
Awareness ceiling	0.98	maximum awareness

Safety-Adjusted Demand Modifier

The serious adverse event rate adjusts the effective quality score:

Serious AE Rate	Quality Multiplier	Interpretation
> 5%	0.5x	"Experimental, risky" -- halves pe
3-5%	0.7x	"Promising but dangerous"
1-3%	1.0x	"Acceptable for high-value therapy
0.5-1%	1.3x	"Safe enough for broader adoption"
< 0.5%	2.0x	"Routine medical procedure"
< 0.1%	3.0x	"Over-the-counter safe"

This multiplier is applied to the quality_score before entering the logit:

```
effective_quality_i = quality_score_i * ae_demand_modifier(serious_ae_rate_i)
```

Quality Composite Score

```
quality_score = w_eff(t) * efficacy_index + w_saf(t) * safety_index + w_con(t) * convenience_index
```

Component Indices (0-100 scale)

Component	Gen 1	Gen 2	Gen 3	Gen 4	Calculation
Efficacy index	35	55	75	90	4 * epigenetic_ag
Safety index	27	75	95	98	100 * (1 - seriou
Convenience index	20	50	80	95	Lookup by deliver

Time-Varying Weights

Period	w_eff	w_saf	w_con	Rationale
Years 1-3 (Q1-Q12)	0.50	0.30	0.20	Early adopters care
Years 4-7 (Q13-Q28)	0.35	0.40	0.25	Safety becomes key a
Years 8-12 (Q29-Q48)	0.25	0.40	0.35	Convenience matters
Years 13-20 (Q49-Q80	0.20	0.35	0.45	Mass market wants ea

Weights transition linearly between periods (no discontinuity).

Capability, Brand, and Capacity Stocks

Capability Stock (A_it) -- Drives Efficacy/Safety

```
A_t = (1 - delta_A) * A_{t-1} + eta_A * actual_product_rd_spend
```

Parameter	Symbol	Value	Notes
Depreciation rate	delta_A	0.025/quarter	10% annual; knowledge dec
Accumulation rate	eta_A	0.8 per $1M	$30M/quarter -> +24 point
Starting value (all firms	A_0	35.0	Baseline Gen 1 quality
Gen 2 threshold	--	120.0	Stochastic check begins a
Gen 3 threshold	--	280.0	
Gen 4 threshold	--	500.0	

When the stochastic check triggers a generation advance, the efficacy/safety indices jump to the new generation's values (see table above). The capability stock continues to accumulate beyond the threshold, improving efficacy/safety incrementally within a generation.

Brand Stock (B_it) -- Drives Brand Utility

```
B_t = (1 - delta_B) * B_{t-1} + eta_B * actual_sga_spend * quality_effectiveness
```

Parameter	Symbol	Value	Notes
Depreciation rate	delta_B	0.10/quarter	40% annual; brand fades f
Accumulation rate	eta_B	1.5 per $1M	$10M/quarter -> +15 point
Starting value	B_0	10.0	Low initial brand
Quality effectiveness	--	effective_quality / 50	Marketing more effective
Brand crash on safety eve	--	-30% of current B	Per publicized AE event a

Capacity Stock (K_it) -- Drives Max Production

```
K_effective_t = K_installed_t * (1 - aging_penalty_t)
```

Parameter	Value	Notes
Starting capacity	250 courses/quarter	Pilot plant
Capacity per $1M capex	~4.3 courses/quarter	At small-commercial scale ($120M -
Build delay	4-8 quarters	Depends on scale
Depreciation (PPE)	2.5%/quarter of gross PPE	10% annual
Aging penalty (no reinvestment)	+0.5%/quarter after 20Q	Equipment wear
Minimum utilization to avoid cost	70%	Below this, COGS multiplier kicks

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Cost Structure

COGS Computation (Per Treatment Course)

```
base_cogs = generation_base_cogs * (1 - process_rd_reduction)
utilization_cogs = base_cogs * utilization_multiplier(capacity_utilization)
final_cogs = utilization_cogs
```

Generation	Base COGS/course	At Scale
Gen 1	$14,000	$13,600 at 5000+ courses/year
Gen 2	$7,500	
Gen 3	$2,500	
Gen 4	$800	

Note: The $14,000 base includes per-firm variation of +/-$1,000 drawn at firm creation. This is why the dossier example shows $14,200 while the world doc baseline is $13,600 (which assumes full commercial scale + process optimization).

Capacity Utilization Multiplier

Utilization	COGS Multiplier	Formula
>= 90%	1.00	--
70-90%	1.00 + 0.5*(0.90 - util)	Linear interpolation
50-70%	1.10 + 1.0*(0.70 - util)	Steeper slope
30-50%	1.30 + 1.5*(0.50 - util)	Even steeper
< 30%	1.60 + 2.0*(0.30 - util)	Severe penalty

This multiplier is applied to base_cogs when computing actual COGS posted to the income statement. It captures the fixed-cost absorption effect: when a factory runs below capacity, fixed costs are spread over fewer units.

Workforce Parameters

Workforce Mechanics

Workforce size affects operations through three channels:

```
1. R&D speed: More scientists -> faster capability accumulation
   effective_eta_A = eta_A * (1 + 0.3 * log(scientists / 50))
   At 50 scientists (baseline): multiplier = 1.0x
   At 100 scientists: multiplier = 1.21x
   At 200 scientists: multiplier = 1.42x
```

```
2. Manufacturing quality: More ops staff -> lower batch failure rate
   batch_failure_rate = base_rate * (80 / ops_staff)^0.3
   At 80 ops staff (baseline): 1.0x base rate
   At 120 ops staff: 0.88x base rate (fewer failures)
   At 50 ops staff: 1.15x base rate (more failures)

3. Commercial effectiveness: More sales staff -> higher brand accumulation
   effective_eta_B = eta_B * (1 + 0.2 * log(sales_staff / 30))
```

Workforce Cost Model

Role Category	Base Annual Salary	Overhead Multiplier	Total Cost/Employee/Quart
Research scientists	$185,000	1.4x (benefits, equipment	$64,750
Process development	$155,000	1.3x	$50,375
Manufacturing ops	$95,000	1.25x	$29,688
Quality assurance	$120,000	1.3x	$39,000
Commercial/sales	$140,000	1.5x (commissions, travel	$52,500
Medical affairs	$165,000	1.3x	$53,625
General & admin	$130,000	1.2x	$39,000

Workforce costs flow into SGA (commercial, G&A), R&D (scientists, process dev), and COGS (manufacturing, QA) proportionally.

Workforce Decisions and Effects

Decision	Effect	Lag
Hire N employees (by category)	+headcount, +cost, +capability aft	1 quarter ramp-up (50% effectivene
Lay off N employees	-headcount, -cost, restructuring c	Immediate cost; capacity reduction
Increase compensation X%	Reduces turnover, improves retenti	Immediate cost; retention benefit
Freeze hiring	No new hires; natural attrition co	Gradual headcount decline

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Leasing and Facility Decisions

Lease vs. Build

Firms can choose to LEASE manufacturing capacity instead of building:

Decision	Lease	Build (Capex)
Upfront cost	$0	Full build cost (e.g., $120M for s
Quarterly cost	$8-12M/quarter (lease payment)	$2M/quarter (maintenance)
Capacity available	Immediately (next quarter)	After build delay (4-8 quarters)
Balance sheet	Operating lease liability (loq)	PPE asset (ppentq)
Flexibility	Can terminate with 4Q notice	Permanent (can idle but not sell)
Customization	Standard facility	Fully customized
Capacity	Fixed	Can expand later

Lease Accounting

Under the simulation's accounting rules (simplified ASC 842):

- Right-of-use asset: Recorded as part of PPE (ppentq), equal to present value of remaining lease payments

• Lease liability: Recorded as part of other long-term liabilities (loq)

• Lease expense: Recorded as part of COGS (for manufacturing leases) or SGA (for office leases)

• Depreciation: Right-of-use asset is amortized over the lease term

Compustat Columns for Leases

Column	Name	Source
rouq	Right-of-use assets	PV of remaining lease payments
leaseq	Lease liability (total)	Same as rouq at inception; diverge
xlrq	Lease expense (quarterly)	Lease payment allocated to P&L

Stock-Based Compensation

How It Works

Firms grant stock options and restricted stock units (RSUs) to employees as part of compensation. This is especially important in biotech where cash is scarce.

Firm Decision

Each quarter, firms decide:

```
{
  "stock_comp": {
    "option_grants": 100000,          // number of options granted
    "option_strike_price": "at_market", // at current stock price
    "rsu_grants": 50000,               // number of RSUs granted
    "vesting_quarters": 12             // vest over 12 quarters
  }
}
```

Accounting Treatment

• Expense: Stock-based compensation is a non-cash SGA expense recognized over the vesting period

• Fair value: Options valued using Black-Scholes at grant date; RSUs valued at grant-date stock price

• P&L impact: Reduces operating income (non-cash)

• Cash flow: Added back to CFO (non-cash expense)

• Dilution: Options/RSUs increase diluted share count when exercised/vested

• EPS impact: Diluted EPS uses higher share count

Parameters

Parameter	Value
Typical option grant (% of shares outstanding)	0.5-2.0% per year
Typical RSU grant (% of shares outstanding)	0.3-1.0% per year
Option vesting period	12 quarters (3 years)
RSU vesting period	8-16 quarters
Option exercise ratio (employees who exercise)	70-85%

Black-Scholes volatility assumption	40-60% (biotech)
Expected option life	2.5 years

Compustat Columns for Stock Compensation

Column	Name	Source
stkcpq	Stock-based compensation expense	Fair value of grants / vesting per
diluted_shares	Diluted shares outstanding	Basic + in-the-money options + unv

Tax Parameters

Parameter	Value	Notes
Corporate tax rate	21%	Flat rate, matching US federal
NOL carryforward usage limit	80% of taxable income	Per US TCJA rules
NOL expiration	None (infinite carryforward)	Simplified from real 20-year limit
State taxes	0% (not modeled)	Simplification
R&D tax credit	0% (not modeled)	Simplification; could add as regim
Withholding on dividends	0%	Simplification
Deferred tax asset from NOL	tax_rate * cumulative_NOL	Recorded on BS as part of other as

Financial Institution Parameters

Starting Capital (Canonical Values)

Institution	Starting Capital	Min Capital Ratio	Max Single Exposure
Investment Bank	$5,000,000,000	5% of AUM	N/A (equity, not credit)
Commercial Bank	$2,000,000,000	8% of total commitments	25% of capital
Credit Fund	$3,000,000,000	10% of total deployed	20% of capital

Income and Expense

Parameter	Investment Bank	Commercial Bank	Credit Fund
Fee income	5% of IPO/secondary proce	0.5%/quarter of commitmen	0.3%/quarter of deployed
Operating expense	1% of capital annually	1% of capital annually	1% of capital annually
Interest income	N/A	Revolver balance * rate	Term debt balance * rate

Distress Thresholds (Canonical)

Capital Ratio	Status	Constraint
> 2x minimum	Well-capitalized	None
1x - 2x minimum	Adequate	Mild caution
0.5x - 1x minimum	Undercapitalized	Cannot increase commitments; rate
0 - 0.5x minimum	Critically undercapitalized	No new business; max rates
<= 0	Failed	Replaced next quarter

Macro Shock Parameters

Market Size AR(1) Process

```
log(M_t) = mu + rho * log(M_{t-1}) + sigma * eps_t
```

Parameter	Symbol	Value
Drift	mu	0.005
Persistence	rho	0.95
Volatility	sigma	0.05

Risk-Free Rate Process

```
r_t = max(0.005, min(0.05, r_{t-1} + sigma_r * eps_t))
```

Parameter	Value
Starting value	0.01/quarter (4% annual)
Volatility	0.002/quarter
Floor	0.005/quarter (2% annual)
Ceiling	0.05/quarter (20% annual)

Event Probabilities (Per Quarter)

Event	Per-Firm Probability	Industry Probability
Publicized adverse event	3%	~14% (at least one firm)
Partial clinical hold	2% (higher if AE rate above avg)	--
Full clinical hold	0.5%	--
Class-wide clinical hold	--	0.3%
Academic breakthrough	--	5%
Supply chain disruption	2% per firm	--
Regulatory pricing event	--	~0.5% (~15% cumulative over 80Q)
Insurance coverage mandate	--	~0.4% (~25% cumulative over 80Q)

Stochastic R&D Success

Generation Advance Check

Each quarter, for each firm, if cumulative product R&D exceeds the minimum threshold:

```
P(success) = base_rate * min(2.0, (cumulative / threshold_mid)^0.5)
```

Transition	Threshold (min)	Threshold (mid)	Base Rate	Max P/quarter
Gen 1 -> Gen 2	$400M	$500M	0.12	0.24
Gen 2 -> Gen 3	$800M	$1,000M	0.08	0.16
Gen 3 -> Gen 4	$1,500M	$2,000M	0.06	0.12

"Threshold (min)" is the absolute minimum cumulative spend to have any chance. "Threshold (mid)" is the calibrated center of the range. Spending beyond mid improves probability but with diminishing returns (the square root).

Process R&D (COGS Reduction)

```
cogs_reduction_pct = 0.22 * (1 - exp(-cumulative_process_rd / 120_000_000))
```

This is an exponential saturation: fast gains initially, diminishing returns, maximum 22% reduction within a generation.

Delivery R&D

Same stochastic check as product R&D, with separate thresholds:

Transition	Threshold (min)	Base Rate	Requires
IV -> Subcutaneous	$100M	0.15	Nothing (can do with Gen
SubQ -> Oral	$300M	0.10	Gen 2 compound
Oral -> Gene therapy	$800M	0.06	Gen 3 compound

Scoring Parameters

Equity IRR Computation

Terminal value for surviving firms at end of simulation:

```
terminal_equity_value = equity_price_final * shares_outstanding
```

For defaulted firms: terminal equity = 0 (or waterfall residual, usually 0). For acquired firms: terminal equity = acquisition proceeds to equity holders.

Cash flows to equity: [-IPO investment, +dividends, +buybacks, +terminal_value] IRR computed on this cash flow stream.

This is the canonical formula. Doc 11's "terminal enterprise value" is a separate metric for reporting, NOT used in the IRR calculation.

Environment Rating Scale

Score	Meaning
1-2	Unrealistic, inconsistent, unfair
3-4	Below average; notable problems
5-6	Adequate; functional but some issues
7-8	Good; realistic and engaging
9-10	Excellent; highly immersive and consistent